

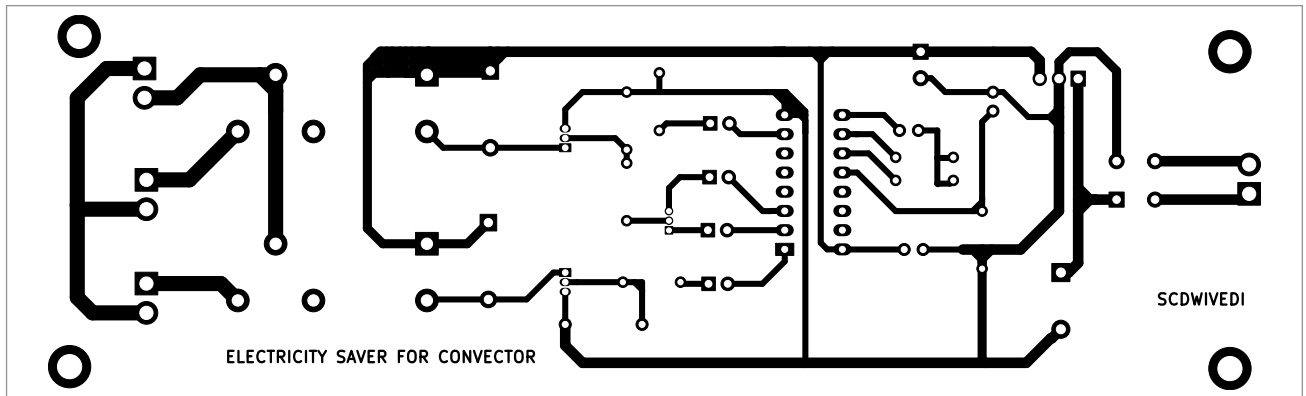


Fig. 3: Actual-size PCB layout

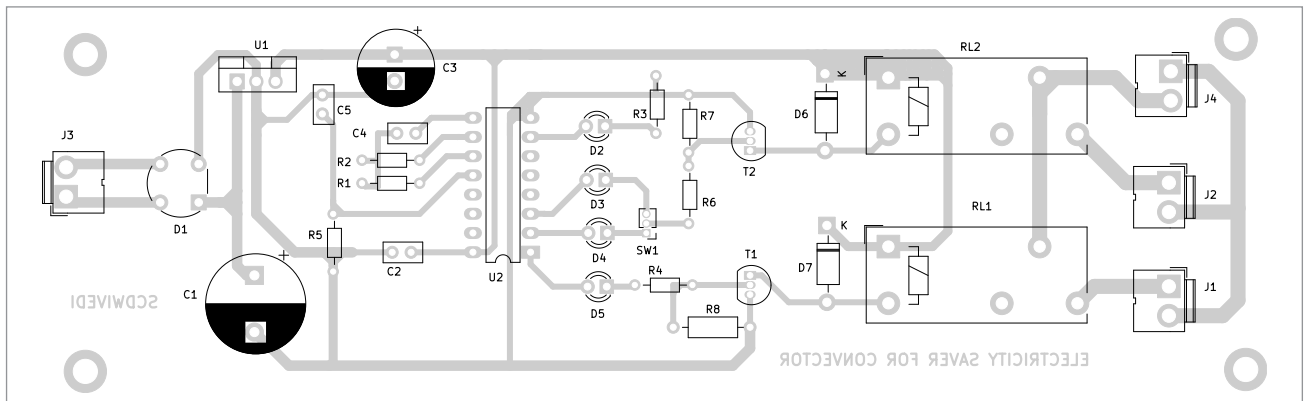


Fig. 4: Component layout of the PCB

connected to the base of T1. Pin 1 will be high for 16 minutes and then low for 16 minutes, and the cycle will keep repeating. Accordingly, HC1 will be on for 16 minutes and off for 16 minutes. Glowing of LED1 means HC1 is on.

Similarly, output pins 2 and 3 are used to connect HC2 via switch SW1 and relay driver transistor T2. LED2 and LED3 are connected to the base of T1 via series resistors R5 and R6, respectively. Pin 2 goes high for 32 minutes and low for next 32 minutes, provided switch SW1 is at position 1. Thereafter the cycle repeats. Accordingly, heat convector HC2 is on for 32 minutes and off for 32 minutes. Glowing of LED2 means HC2 is on.

Pin 3 goes high for one hour and low for next hour, when switch SW1 is at position 3. Thereafter the cycle repeats. Accordingly, heat convector HC2 is on for an hour and off for the next one hour. Glowing of LED3 means HC2 is off.

The time periods are governed by the relationship:

The basic frequency

$$f_{osc} = 1/2.5 (R1 \times C4) \text{ in Hertz}$$

Where R1 is the resistance at pin 10 in ohms and C4 the capacitance

PARTS LIST

Semiconductors:

- U1 - 7812, 12V voltage regulator
- U2 - CD4060 oscillator/counter
- T1, T2 - BC547 NPN transistor
- D1 - Bridge rectifier
- D2-D5 - 5mm LED
- D6 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 1-mega-ohm
- R3, R7 - 10-kilo-ohm
- R4, R6 - 1-kilo-ohm
- R5 - 82-kilo-ohm
- R8 - 47-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2, C5 - 0.1 μ F ceramic disk
- C3 - 10 μ F, 35V electrolytic
- C4 - 0.2 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12-0-12V, 500mA secondary transformer
- SW1 - Toggle switch
- RL1, RL2 - 12V single changeover (1CO) relay
- J1-J4 - 2-pin header connector
- HC1, HC2 - 1.5kW heat convector
- Jumper wires

at pin 9 in Farads. For example, if R1 is 1-mega-ohm and C4 is 0.22 μ F, $f_{osc} = 1/2.5 (1,000,000 \times 0.000,000,22) = 1.8\text{Hz}$. If the selected output is Q3, then $2^n = 2^3 = 8$. Therefore, time period (in seconds) is $t = 2^n / 1.8\text{Hz} = 8 / 1.8 = 4.4$ seconds.

When both HC1 and HC2 are on for 12 hours using this circuit, the convectors will be on for only half the time, that is for just 6 hours each day. So, the power consumed by HC1 and HC2 per day will be just half.

Construction and testing

An actual-size PCB layout for the energy saver for heat convectors is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable plastic box. The connectors for HC1 and HC2 convectors should be fixed on rear side of the box. **EFY**

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